Problem 3

Determine the strength of the Newtonian gravitational field of a body with mass $m = 1$ kg at a distance of $r = 10$ m.

Solution

The strength of the gravitational field is given by the formula for gravitational field intensity:

$$G = \frac{\gamma m}{r^2}$$

Where:

- $\gamma = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$ is the gravitational constant,
- $m = 1$ kg is the mass of the body,
- $r = 10$ m is the distance from the mass.

Substitute the values:

$$G = \frac{6.67 \times 10^{-11} \times 1}{(10)^2}$$

Simplify:

$$G = \frac{6.67 \times 10^{-11}}{100}$$

$$G = 6.67 \times 10^{-13}\ \mathrm{N/kg}$$

Conclusion: The strength of the Newtonian gravitational field at a distance of 10 m from a 1 kg mass is approximately 6.67×10^{-13} N/kg.